EECS 563 Homework 1 Solution

Due: Friday, September 11, 2026, 11:59 pm. Online submission on Canvas.

Problem 1 [9 points]: In class, we talked about various access network technologies.

(a) Name two different types of Access Networks. [3 points]

Solution: Ethernet, Cellular, Wi-Fi, Satellite.

(b) **[Wireless access networks.]** We know that IEEE 802.11 is the technical standard name for wireless local area networks (WLAN), which has been developed by the Institute of Electrical and Electronics Engineers (IEEE). On the other hand, "Wi-Fi" is the marketing name for products that comply with the 802.11 standards. Briefly research and explore the evolution of Wi-Fi technologies and the IEEE 802.11 family of standards (e.g., 802.11a, 802.11b, 802.11n, 802.11ac, 802.11ax). Highlight some key features of each generation. [3 points]

Solution: Answers vary but below are some of the key features in each generation.

- IEEE 802.11 is the first 2.4 GHz Wi-Fi standard introduced in 1997 and is the base for Wi-Fi wireless networks.
- IEEE 802.11b, also referred to as Wi-Fi 1, was introduced to the market in 1999. It also operated at 2.4 GHz, but to reduce interference and achieve higher data rates, it incorporated modulation schemes called direct-sequence spread spectrum/complementary code keying (DSSS/CCK). Wi-Fi 1 enabled wireless communications at distances of 40m indoors and 150m outdoors.
- IEEE 802.11a, or Wi-Fi 2, introduced in 1999, was the successor to IEEE 802.11b. *It was the first Wi-Fi specification to feature a multi-carrier modulation scheme (OFDM) to support high data rates*, unlike Wi-Fi 1's single-carrier design. It supported 5 GHz operation and its 20 MHz bandwidth supports multiple data rate options.
- IEEE 802.11n, or Wi-Fi 4, was introduced in 2009 to support both 2.4 GHz and 5GHz frequency bands. IEEE 802.11n supports up to 600 Mbit/s data rates, multiple channels within each frequency band, and other features.
- IEEE 802.11ac, or Wi-Fi 5, was introduced in 2013 to support data rates up to 3.5 Gbit/s. It was the first Wi-Fi standard to enable the use of multiple input/multiple output (MIMO) technology so that multiple antennas could be used on both sending and receiving devices.
- IEEE 802.11ax, or Wi-Fi 6, is a recent standard published in 2021, which provides a theoretical data rate up to 9.6 Gbit/s.

References:

1) `https://standards.ieee.org/beyond-standards/the-evolution-of-wi-fi-technology-and-standards/`

(c) **[Wired vs. Wireless access networks]** Based on your answer to the previous part and the access network technologies discussed in class, give two differences between

wired and wireless links. For example, you can think about the distance, reliability, and data rates that each type of link provides. [3 points]

Solution: Answers vary. Below are a few key differences between wired and wireless networks. Wired and wireless networks are two common ways to connect devices and enable communication between them. A wired access network uses physical cables to connect devices to a network, for example, a system connected to KU network using an ethernet cable. Wired networks offer higher stability, reliable data rates, and are more secure in general. In wireless access networks, devices transmit data through radio waves such as Wi-Fi or cellular networks. Wireless networks allow for greater mobility but potentially experience interference and, hence may not provide reliable data rates always.

Problem 2 [6 points]: In class, we talked about three main components of the network, namely: Network Edge, Access Network, and Network Core. In this question, we are focusing on Network Edge and Network Core components.

(a) Compare and contrast circuit switching networks and packet switching networks. Describe the advantages and disadvantages of each network. [3 points]

Solution: Circuit switching involves establishing an end-to-end communication path between the source and destination, where this path remains dedicated to the communication for the duration of the session. The table below highlights a few advantages and disadvantages of Circuit switching networks.

Advantages	**Disadvantages**
Guaranteed bandwidth for the duration of the call.	Inefficient use of bandwidth.
Low latency due to predetermined path.	Limited scalability.
Better delay performance guarantees.	High cost due to dedicated resources.

Packet switching is a method of communication where the data is broken into small pieces of variable length and transmitted over the network and routes the packets dynamically. The table below highlights a few advantages and disadvantages of Packet switching networks.

Advantages	**Disadvantages**
Efficient use of bandwidth as the resources are shared.	Higher latency due to dynamic routing of packets.
Highly scalable to handle large amounts of data.	Packet losses due to congestion.
Less expensive because the connections are not permanent.	No delay performance guarantees

(b) Compare Network Core and Network Edge. Give some examples of Network Core and Network Edge devices. [3 points]

Solution: The network core consists of switches and routers that route incoming packets to their next destination. Meanwhile, the network edge consists of users who receive and generate packets. Network core devices are routers of ISP's, while network edge devices are clients and servers.

Problem 3 [6 points]: Layering structure and Internet protocols.

(a) We discussed that the Internet and networks are built on a layered architecture, where each layer performs specific functions. Why is a layered approach used in network design? Explain the benefits of having distinct layers rather than a single unified system. Which of the network layers does an internet user interact with the most? [4 points]

Solution: A layered approach in network design simplifies communication by dividing it into distinct, manageable layers, each with a specific function. Thus layered architecture enhances modularity, flexibility, and scalability. Moreover, the layered design helps in identifying and troubleshooting potential issues quicker than in a single unified system. An internet user primarily interacts with the Application layer, which includes web browsers and other applications they use to access the internet.

(b) Briefly justify that why standardized protocols are essential for Internet communication. [2 points]

Solution: Standardized protocols establish a common set of rules and conventions that allow diverse devices and systems to communicate with each other seamlessly, regardless of the manufacturer or the operating system. These protocols ensure that data is exchanged between the devices securely and efficiently, ensuring global connectivity for the growing number of Internet-connected devices.

Problem 4 [15 points]: Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

(a) Express the propagation delay, d_{prop}, in terms of m and s. [2 points]

Solution: $d_{\text{prop}} = \dfrac{m}{s}$

(b) Determine the transmission time of the packet, d_{trans}, in terms of L and R. [2 points]

Solution: $d_{\text{trans}} = \dfrac{L}{R}$

(c) Ignoring processing and queuing delays, obtain an expression for the end-to-end delay. [2 points]

Solution: $d_{\text{end-to-end}} = d_{\text{trans}} + d_{\text{prop}} = \dfrac{L}{R} + \dfrac{m}{s}$

(d) Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet? [2 points]

Solution: At $t = d_{\text{trans}}$, the last bit has just been pushed onto the link; it is leaving Host A (at Host A's end of the link).

(e) Suppose d_{prop} is greater than d_{trans}. At time $t = d_{\text{trans}}$, where is the first bit of the packet? [2 points]

Solution: The first bit is still in transit on the link and has not yet reached Host B. It has traveled a distance $s \cdot d_{\text{trans}} = \dfrac{sL}{R}$ from Host A, which is less than m since $d_{\text{prop}} > d_{\text{trans}}$.

(f) Suppose d_{prop} is less than d_{trans}. At time $t = d_{\text{trans}}$, where is the first bit of the packet? [2 points]

Solution: The first bit has already arrived at Host B (it reached B at time $d_{\text{prop}} < d_{\text{trans}}$).

(g) Suppose $s = 2.5 \times 10^8$ m/sec, $L = 120$ bits, and $R = 56$ kbps. Find the distance m so that d_{prop} equals d_{trans}. [3 points]

Solution: Set $\dfrac{m}{s} = \dfrac{L}{R}$, so

$$m = \frac{Ls}{R} = \frac{120 \times 2.5 \times 10^8}{56 \times 10^3} \approx 5.36 \times 10^5 \text{ m} \approx 536 \text{ km}.$$

Problem 5 [12 points]: Consider the following network scenario. Suppose multiple users share a 20 Mbps link, and each user transmits at the rate of 2 Mbps when "active". Each user is active 15 percent of the time.

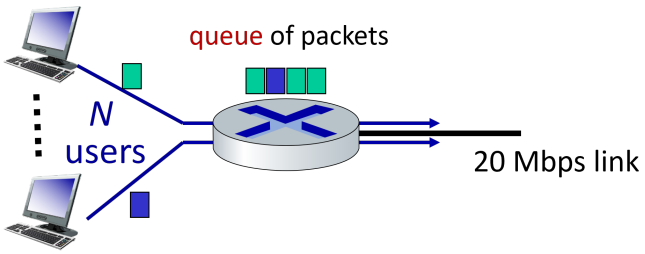


Figure 1: Network topology for Problem 5.

(a) When circuit switching is used, how many users can be supported? [2 points]

Solution: Circuit switching reserves a dedicated 2 Mbps for each user, so the link supports $\dfrac{20 \text{ Mbps}}{2 \text{ Mbps}} = 10$ users.

(b) For the remainder of this problem, suppose packet switching is used with $N = 30$ users. Find the probability that only one user is transmitting [3 points].

Solution: With $p = 0.15$ and $N = 30$, the number transmitting is Binomial$(30, 0.15)$.

$$P(\text{exactly } 1) = \binom{30}{1}(0.15)(0.85)^{29} \approx 0.0404.$$

(c) Find the probability that all 30 users transmit at the same time. [3 points]

Solution:

$$P(\text{all } 30) = (0.15)^{30} \approx 1.92 \times 10^{-25}.$$

(d) Find the probability that there are 11 or more users transmitting simultaneously. [4 points]

Solution: The link can carry 10 simultaneous users; 11 or more causes congestion.

$$P(\geq 11) = \sum_{k=11}^{30} \binom{30}{k} (0.15)^k (0.85)^{30-k} \approx 0.00294.$$

Solution(alt): The event "11 or more transmitting" is the complement of "10 or fewer transmitting":

$$P(\geq 11) = 1 - P(\leq 10) = 1 - \sum_{k=0}^{10} \binom{30}{k} (0.15)^k (0.85)^{30-k}.$$

Evaluating the sum over $k = 0, 1, \ldots, 10$ gives $P(\leq 10) \approx 0.99706$. Therefore

$$P(\geq 11) = 1 - 0.99706 \approx 0.00294.$$

Problem 6 [6 points]: Suppose you would like to urgently deliver 40 terabytes (1 TB = 10^{12} bytes) data from Boston to Los Angeles. You have available a 100 Mbps dedicated link for data transfer. Would you prefer to transmit the data via this link or instead use FedEx overnight delivery? Briefly justify your answer.

Solution: Data size in bits: $40 \times 10^{12} \times 8 = 3.2 \times 10^{14}$ bits.

Just considering the transmission delay over this link (and ignoring other types of delay), we get:

$$t = \frac{3.2 \times 10^{14} \text{ bits}}{100 \times 10^6 \text{ bps}} = 3.2 \times 10^6 \text{ s} \approx 889 \text{ hours} \approx 37 \text{ days}.$$

Since the link takes about 37 days versus FedEx's overnight ($\sim$ 1 day) delivery, it is preferred to use **FedEx**.

Problem 7 [16 points]: This question asks you to perform `traceroute` to the host `stanford.edu` using the Kali Linux image that you have installed. You can use the following command.

```
sudo traceroute -I stanford.edu
```

For each of the following parts, **please include a screenshot of both the command and its output.**

(a) What is the distance (in number of hops) between your host and stanford.edu? Did you reach stanford.edu? if not, count and report the distance between your host and the last router inside the Stanford network. [3 points]
Solution: As shown in the Fig 2, the distance between the host and stanford.edu is 20 hops. Your answer could vary depending on the day, time and the location where you will be performing the experiment.

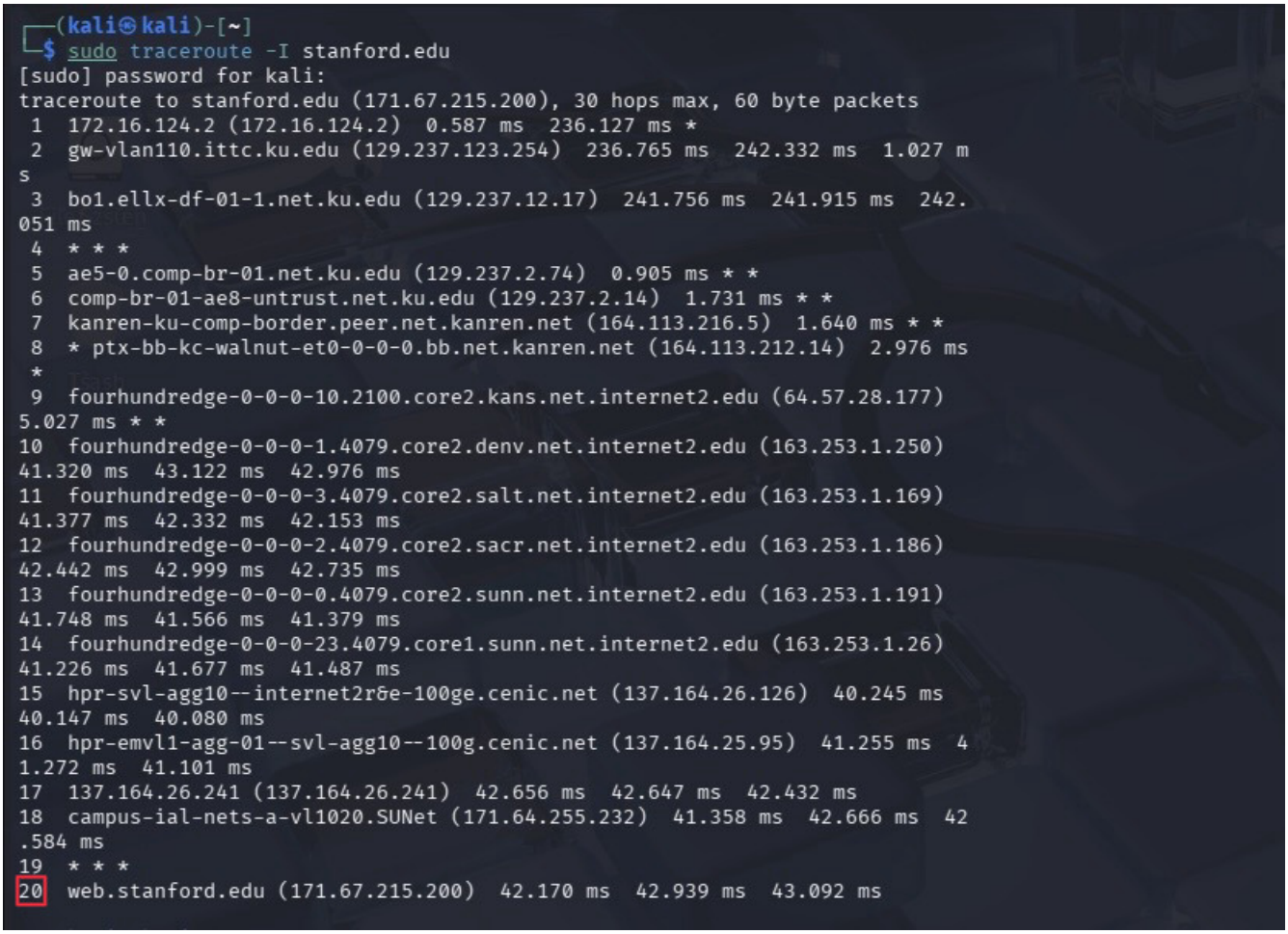


Figure 2: Traceroute to stanford.edu

(b) Perform a `traceroute` command to the host stanford.edu, but this time 5 probe queries should be sent rather than just three (hint: run the command `man traceroute` to find out the correct option). Give the syntax of the `traceroute` command used to send 5 queries. [3 points]
Solution: `sudo traceroute -I -q 5 stanford.edu`. To perform 5 probes, we add the option "-q 5" to the traceroute command used earlier. As shown in Fig. 3, we see 5 entries on each hop. "*" implies the packet/ probe didn't reach the destination before the default timer expired.

(c) Repeat the above experiment at two different times of the day (e.g., morning vs. afternoon) and from two different access networks (e.g., KU WiFi network, cellular hotspot

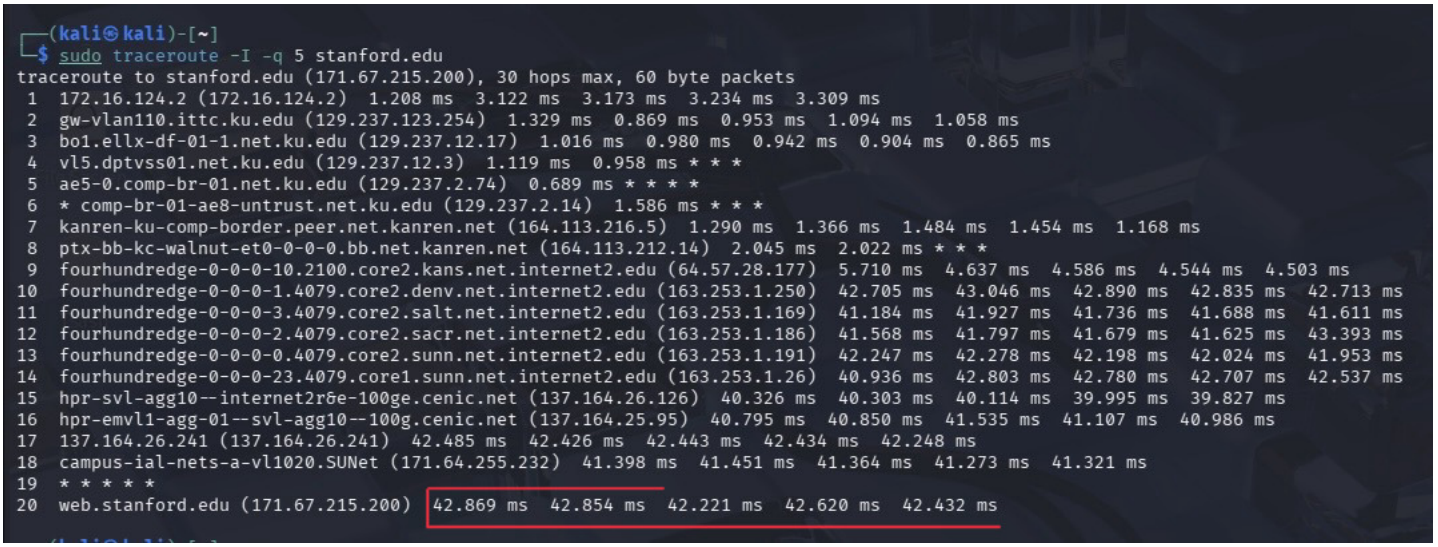


Figure 3: Traceroute to stanford.edu with 5 probes

network, wired network, etc.). Report your observations. Do you observe any differences in the number of hops and delay values? [4 points]

Solution: Answers would vary, but you should see a difference for delay values at different times of the day, as the level of underlying traffic can change at different times with respective to the network.

(d) Your traceroute output in part (a) likely passed through routers with names containing `kanren.net` and `internet2.edu` before reaching Stanford. Briefly research the Internet2 network and answer the following, in no more than one paragraph [6 points].

- What is Internet2? who operates it, who its members are, and what purpose it serves.
- Where does KanREN fit between KU and Internet2? Use the terms access network, regional network, and backbone network from the "network of networks" model discussed in class.
- Your packets reached Stanford without traversing a commercial tier-1 ISP like AT&T or Verizon. What does this suggest about how the Internet is actually structured?

Solution:

Solution: Internet2 is a U.S. not-for-profit R&E networking consortium (registered as UCAID, based in Ann Arbor, MI), whose members are mainly U.S. universities plus industry partners, regional networks, and affiliates; it operates a high-performance backbone for data-intensive research and education. In the "network of networks" model, KU reaches KanREN (the **regional network**) through KU's campus **access network**, and KanREN connects up to the Internet2 **backbone network** carrying traffic to Stanford. Reaching Stanford without a tier-1 commercial ISP shows the Internet is many independent, interconnected networks rather than one hierarchy.

References:

- Internet2, *About Us*. https://internet2.edu/community/about-us/
- Internet2, *Membership*. https://internet2.edu/community/membership/
- Wikipedia, *Internet2*. https://en.wikipedia.org/wiki/Internet2